Built-In Entertainment Center

Path A - Build & Install Manual

Colored cut guides and step-by-step assembly
Joinery: Kreg pocket holes (primary) | face screws where noted



Design locks: centered 119" carcass group | ~20-1/8" open flanks | side risers 3-1/2" (2x4 on edge) | 3/4" side counters | shelf towers to crown | White Dove | beadboard behind unit only | TV stays wall-mounted.

1. What you are building

Path A wraps three existing cabinets (27" + 65" + 27") into one painted built-in. Side cabinets sit on 2x4 ladder risers (3-1/2" tall). Side counters are 3/4" plywood, stepped about 4-1/4" above the center. Open shelf towers go to the crown. Outer wall flanks stay open (no cabinets to the wall).

Actual lumber sizes

- 2x4 -> actual 1-1/2" x 3-1/2" (on edge = 3-1/2" riser height)
- 3/4" plywood -> 0.75" thick (verify; some sheets are 23/32")
- Side stack AFF: 3-1/2" + 30" cabinet + 3/4" counter = 34-1/4"
- Center console: ~30" AFF (no riser)

Joinery rules

Kreg pocket holes (default)

Use for riser frames and tower carcass joints (sides to top/bottom/shelves). Drill on the hidden face - inside of frames, underside of shelves.

Face / through screws (where it makes sense)

- Counters into cabinet from below.
- Cabinet down into riser rails from inside the cabinet.
- Towers into wall studs (anti-tip).
- Optional face screws through tower sides into shelf edges.

2. Tools & materials

Tools

- Kreg pocket-hole jig, stepped bit, and driver
- Drill/driver, clamps, square, tape, level
- Circular / track saw (or table saw) for plywood
- Miter saw helpful for 2x4s
- Sander, caulk gun, paint supplies

Buy list (carcass work)

- 2x4 x 8' - about 3 sticks (both risers)
- 3/4" plywood 4x8 - about 3 sheets (towers + counters; more later for face skins)
- Kreg screws: 1-1/4" for 3/4" plywood, longer for 2x4 (use jig chart)
- Wood glue; 1-1/4" to 2" wood screws for face/through fastening
- Edge banding, primer, Benjamin Moore White Dove
- Beadboard for upper center bay only

Kreg settings (typical)

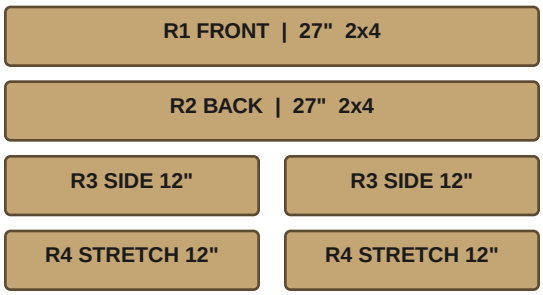
- 3/4" plywood -> jig set to 3/4"
- 2x4 (1-1/2") -> jig set to 1-1/2"

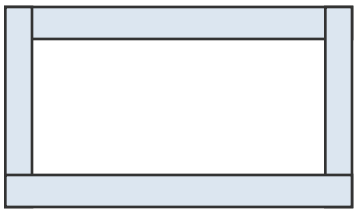
Confirm screw length with your jig chart - brands vary slightly.

3. Cut guides - board maps

3A - Side risers (make 2)

Outer footprint 27" W x 15" D x 3-1/2" H. Front/back rails 27"; sides and stretchers 12" (15 - 1-1/2 - 1-1/2).

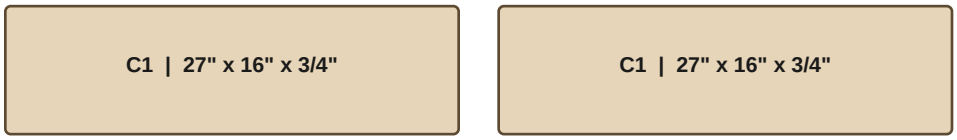




Plan 27" x 15"

Both risers together: 4x 27" and 8x 12" (~3 sticks of 8' 2x4).

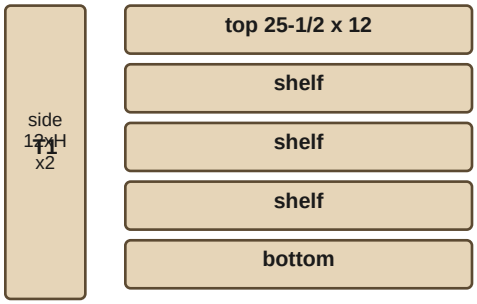
3B - Side counters (3/4" plywood)



Qty 2. 1" front overhang past 15"-deep cabinet. Edge-band exposed edges. Center stays lower.

3C - Shelf towers (3/4" plywood)

Outer 27" x 12". Horizontals between sides = 25-1/2" x 12". Height H = (floor to crown underside) - 34-1/4". Placeholder ~58-1/4" - measure before cutting sides.



Per tower: 2 sides + top + bottom + 4 shelves.
Both towers: 4 sides + 12 horizontals.

4. Step-by-step install

1. **Measure crown (before tower cuts)**
   At left and right tower locations, measure floor to underside of crown. H = that measurement - 34-1/4". Write H on the cut list before cutting tower sides.

2. **Cut and pocket-screw the risers**
   Cut 2x4 parts. Orient on edge (3-1/2" tall). Dry-fit a 27x15 rectangle. Pocket-hole the 12" sides and stretchers into the front/back rails from the inside. Glue, clamp, drive Kreg screws. Check square. Build two identical risers.

3. **Place risers and side cabinets**
   Set risers on the floor with the 65" center console as one 119" group (~20-1/8" open flank each side). Level. Set side cabinets on risers. From inside each cabinet, drive face screws down into the 2x4 rails (pilot holes).

4. **Dry-fit the three units**
   Keep only 1/8"-1/2" scribe gaps between units (no 6" fillers). Confirm open flanks. Mark the wall for beadboard in the upper center bay only. Keep the TV as mounted.

5. **Cut and install side counters**
   Cut two 27x16x3/4" blanks. Edge-band. Set with ~1" front overhang. From below, use face screws up into the plywood. Side tops should sit ~4-1/4" above the center top.

4. Install (continued)

6. **Build shelf tower boxes**
   Cut sides to measured H and horizontals to 25-1/2 x 12. Pocket-hole top, bottom, and shelves into the side panels from the underside/inside. Keep boxes square. Space shelves evenly (or use pins for adjustable shelves).

7. **Set towers on side counters**
   Align towers with the cabinets below. From inside the lowest bay, face-screw down through the tower bottom into the counter. Plumb each tower.

8. **Anchor towers to the wall**
   Find studs. Use face/through screws into studs (through the tower side or a French cleat) for anti-tip. Prefer structural screws into studs over pocket holes alone.

9. **Beadboard + face skins**
   Install beadboard on the wall between the towers only. Add plywood face skins/returns so the three units read as one White Dove built-in. Hide fasteners where possible; fill and paint.

10. **Finish**
    Plug or fill pocket holes, sand, prime, paint White Dove. Caulk scribes. Reinstall doors. Route power to cabinet tops; plan outlet relocation behind the TV per local code.

Weekend order

- Day 1: measure crown | cut/assemble risers | place cabinets | counters
- Day 2: build/set towers | anchor | start beadboard / skins

5. Joinery quick reference

Use pocket holes for

- 2x4 riser: ends of 12" parts into front & back rails (inside face)
- Tower: top, bottom, shelves into side panels (pockets on underside)
- Later hidden cleats / plywood butt joints

Use face / through screws for

- Cabinet -> riser (inside cabinet, down into rails)
- Counter -> cabinet (from below)
- Tower -> wall studs (anti-tip)
- Tower bottom -> counter (inside lowest bay)

Avoid

- Visible pocket holes on finished White Dove faces (or plug and paint carefully)
- Glue-only structural joints

> **Tip**
> Clamp every pocket joint before driving screws. On plywood, use the screw type your Kreg chart lists for 3/4" stock.

Repo files

- build-guides/01-risers.md | 02-counters.md | 03-shelf-towers.md
- mockups/path-A-build-guide.svg | mockup-path-A-photoreal.jpg
- Cursor canvases: entertainment-center-mockups + path-a-*-cut-guide

Re-check ceiling height before cutting tower sides. Follow local electrical code for any outlet work.